ENG518 Midterm Short Notes BY PIN2 and MUHAMMAD (MAS All Rounder)

Lecture: 01

Research - it urges for the discovery of truth (critical thinking), defining and redefining problems, formulating hypotheses, collecting, organizing and evaluating data, making deductions and reaching conclusions. **Characteristics of research**

- The nature of research is systematic-logical- empirical-reductive-replicable.
- It gathers new knowledge or data from primary or first-hand sources.
- Emphasis upon the discovery of general principles

- Uses certain valid data gathering devices
- Logical and objective
- Eliminates personal feelings and preferences
- Carefully recorded and reported
- Conclusions and generalizations are arrived at carefully and cautiously

Functions of Research

- Refinement and extension of knowledge happens
- Makes decision-making easier
- Improves learning and teaching processes
- Improves various aspects of human life

Lecture: 02

Areas of concern in ELT research

- Students
- Teachers (ES/FL)
- Parents
- Admin
- Stakeholders
- Understanding of language issues **What is ELT Research About?**

- Language policy
- Practical issues
- Problems related to second or foreign language
- The preservation of endangered languages
- Bilingualism
- Multilingualism
- Language education
- Assessment and evaluation
- Treatment of language
- Cross-cultural communication

Lecture: 03

Characteristics of Basic research

- Theoretical
- Contributing to subject knowledge

- No immediate value
- Used for broader studies

Characteristics of Applied research

- Action research
- Solving problems
- Specifying a problem

Types of research on the basis of:

Research objectives

- Fundamental
- Action

Nature of research

- Exploratory
- Confirmatory

Data collection procedures

- Qualitative
- Quantitative

Application of research

- Longitudinal
- Cross-sectional
- Conceptual
- Empirical

Other types of research

- Educational research
- Historical research

Educational research - the issues of influencing educational theories and practices are emphasized.

Consideration in educational researches

- Teacher behavior
- Research having educational implication
- Educational administration
- Educational technology
- Educational significance

Scientific ways to solve problems

- Developing the problem
- Forming the hypothesis

- Gathering the data
- Analyzing and interpreting data
- Conclusion

Characteristics of an Investigator

- Reflective thinking
- Objectivity in thinking
- Scientific attitude
- Democratic behavior
- Sensitive towards job
- Creative and imaginative
- Patience
- Open-minded
- Knowledge of action research

Lecture: 04

Assortment of a Problem

First step in research is the *selection* of a problem. A problem is selected through reflective thinking towards a target issue; a careful ordered thinking is required. After selecting a problem the keen *understanding* and evaluation of the research problem is required. **Reflective and scientific thinking**

- Definition
- Explanation
- Elaboration
- Collection
- Conclusion

Scientific thinking involves:

- Inductive-deductive reasoning
- Cause and effect
- Assumption/hypothesis

Identification of a Problem

Researcher needs to know about the following things:

- Exact nature and dimensions of the problem
- Research field
- Mastery of the area **Sources of a problem**

- Personal experience
- Studying available literature
- Innovation (technological changes)
- Secondary sources; publications/reports **Criteria for the Selection of a Problem**

- Novelty (avoid unnecessary duplication)
- Importance for the field
- Interest intellectual curiosity

- Training and personal qualification
- Availability of data
- Cost and return
- Time factor

Statement of a Problem

A problem statement is the description of an issue currently existing which needs to be addressed. **Three criteria for a good problem statement**

1) Concerned with variables involved
2) Clear and unambiguous
3) Amenable to empirical testing

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Assumption is a realistic expectation which is something that we believe to be true. However, no adequate evidence exists to support this belief.

Postulates - The scientific research method is based on certain basic postulates.

Hypotheses - A hypothesis is a tentative statement about the relationship between two or more variables. It is a specific, testable prediction about what you expect to happen in a study.

Characteristics of Assumptions

- On logical insight
- Truthfulness observed
- Taking things for granted
- Restrictive conditions

Characteristics of Postulates

- For starting working
- Discovery of other things
- Not proven – simply accepted at face value
- Working beliefs of most scientific activity

Characteristics of Hypothesis

- No conflict with universal laws
- Permits deductive reasoning
- Simplest possible form
- Clear verbalization
- Controls for verification
- Effective use of tools/tech

Importance of a hypothesis

- Eyes of investigator

- Focused research
- Clear and specific goals
- Linking together
- Guiding light
- Direction to research
- Stimulating further research **Kinds of a Hypothesis** Basically of two types:

1) Relational 2) Causal Further types:

- Directional
  - ✦ Expected direction
  - ✦ Specific direction
- Null (ND)
  - ✦ Assertion that NO relation exists
  - ✦ Probability framework
  - ✦ More in education and psychology
- With different names (simple/complex/alternative/statistical/logical)
- Declarative
  - ✦ Anticipated relationship
  - ✦ Existing evidence examined
- Question
  - ✦ NO general consensus
  - ✦ Simplest level of empirical evidence
  - ✦ Question may or may not qualify as a hypothesis

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Variables in a Hypothesis

A variable is anything that has a quantity or quality that varies. A hypothesis is made testable by operational definitions of variables and terms.

Types of variables

- **Independent**
  - ✦ Provides a stimulus
  - ✦ Measured
  - ✦ The cause of change
  - ✦ Affect others
- **Dependent**
  - ✦ Responsible for the response/output
  - ✦ Observed to effect
- **Moderator – control – intervening** *Moderator*
  - ✦ Secondary independent variable ✦ Modifying the relation

  Control variable in scientific experimentation is an experimental element which is constant and unchanged throughout the course of the investigation.

Intervening variable is a hypothetical variable used to explain causal links between other variables

Example:

Among students of the same age and intelligence, skill performance is directly related to the number of practice traits particularly among boys but less directly among girls.

IV: Number of practice traits DV:

Skill performance

MD: Sex

CV: Age, intelligence

IntV: Learning

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Preliminary sources are:

- Educational Index
- MLA/APA
- CIJE
- SSI (Social Sci. Index)
- ERIC/RIE

Secondary sources can be:

- Literature reviews/position papers/books
- Tables of references/bibliographies

ERIC - an online library of education research and information, sponsored by the Institute of Education Sciences (IES) of the U.S. Department of Education

HTLR: Preliminary Sources - The resource materials that come from experts can help us find research

- Storehouses
- CD – Rom
- University library
- HEC/other digital resources

HTLR: Secondary Sources

Secondary sources are the references and summaries; at this point primary research is accessed through the eyes of someone other the actual researchers (who did it)

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Use the following options for locating primary research LPR/ research articles:

- ERIC
- JSTOR
- www.eric.ed.gov
- Google Scholar

Position Papers vs. Primary Research

A position paper is a written report outlining someone's attitude or intentions regarding a particular matter. It is similar to literature review.

Primary research is not included. It is used for proposing answers to research questions. It helps in generating hypothesis.

Use the following studies for locating primary research LPR:

- Particular issues
- Benchmark studies
- Use Seminal studies
- Profitable places to find research studies
- Studies with sparking interest of the area

TWO criteria

1. **Refereed** (pass a matter to (a higher body) for a decision)
2. **Blind review** (This means that the reviewers of the paper won't get to know the identity of the author(s), and the author(s) won't get to know the identity of the reviewer.)

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Research journals available to consult for ELT research

- TESOL
- AAAL
- IATEFL
- AERA
- ESP journal

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Population - the entire mass of observation (the universe). All the members of the group /objects are generalized as population. It is impossible to access all the population.

Sample - the data drawn to test hypothesis. It is the data source to answer research questions. Sample contains the characteristics of a specific group.

Sampling Paradigms

Basically, there are two sampling paradigms, depending upon the nature and purpose of the study.

1) Information rich paradigm
2) Representative sampling paradigm

Information rich paradigm

- Uncover the information
- It requires in-depth analysis
- Mainly in qualitative research
- Views – feelings – holistic small sample size

- Less generalizable

Representative sampling paradigm

- Replica of a larger group
- Generalizable
- Large samples (Questionnaire etc.)

Lecture: 11 Components and steps involved in research planning:

- Devise research objectives and sampling
- Research strategy
- Analyzing data
- Reporting the results
- Selecting qualitative & quantitative procedures

Sampling Techniques

1) Probability Sampling
2) Non- Probability Sampling

Probability Sampling

This Sampling technique uses randomization to make sure that every element of the population gets an equal chance to be part of the selected sample. It's alternatively known as random sampling.

- Simple Random Sampling
- Stratified sampling
- Systematic sampling
- Cluster Sampling
- Multi stage Sampling

Non-Probability Sampling

This technique is more reliant on the researcher's ability to select elements for a sample. Outcome of sampling might be biased and makes difficult for all the elements of population to be part of the sample equally. This type of sampling is also known as non-random sampling.

- Convenience Sampling
- Purposive Sampling
- Quota Sampling
- Snowball Sampling

Sampling Designs

A sample design is made up of two elements.

Sampling method - rules and procedures by which some elements of the population are included in the sample. Some common sampling methods are:

- Simple random sampling
- Stratified sampling
- Cluster sampling

Estimator - The estimation process for calculating sample statistics is called the estimator. Different sampling methods may use different estimators. For example, the formula for computing a mean score with a simple random sample is different from the formula for computing a mean score with a stratified sample. **Characteristics of a sample**

- Free from bias
- Objective
- Comprehensive
- Economical
- Approachable
- Accuracy
- Practicability

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Errors in Sampling

A sample is not necessarily always a true representative of population. There can be errors occurred in sampling. Two types of errors:

1) Random error
2) Systematic error

Internal validity - A research study with high internal validity lets you choose one explanation over another with a lot of confidence, because it avoids confounding (more than one possible independent variables acting at the same time). The less chance for confounding in a study, the higher its internal validity is.

External validity refers to how well data and theories from one setting apply to another. It determines the degree of generalizability. Can the findings of the study be generalized to the whole population?

Threats to external validity

- Participant characteristics should be studied keenly
- Relating to the setting of the study
- Timing of the study

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Research Designs

Research design is the blue print of your study, there can be more than one possibilities in selecting research design for the study. The step wise study framework is given below:

Problems (Variables) – **Research Questions – population– sample – Research Designs**

Characteristics of a Research Design

- Research design must be the plan that can guide the strongest plan for your research.
- Research design should be the most efficient structure to provide data for your study.
- Research designs must be the most useful plan to answer research questions.
- A poor research design is a house full of problems. **Type of research design OR CRD: Three Continua**

1. Basic-Applied
2. Qualitative – quantitative
3. Exploratory – confirmatory

Basic/Pure/Fundamental research - It focuses on advancing scientific knowledge for the complete understanding of a topic or certain natural phenomenon. In other words, when knowledge is acquired for the sake of knowledge.

Applied research - It is directed towards real life application or providing a solution to the specific practical problems and develop innovative technology.

Characteristics:

- Action research
- Solving problems
- Practical

Basis for Comparison	Basic Research	Applied Research
Nature	Theoretical	Practical
Utility	Universal	Limited
Concerned with	Developing scientific knowledge and predictions	Development of technology and technique
Goal	To add some knowledge to the existing one.	To find out solution for the problem at hand.

Quantitative research - produces numerical data and hard facts. It aims at establishing cause and effect relationship between two variables by using mathematical, computational and statistical methods. The research is also known as *empirical research* as it can be accurately and precisely measured.

Characteristics:

- Psychology
- Statistics (generalize)
- To larger populations

Qualitative research - used to gain an in-depth understanding of human behavior, experience, attitudes, intentions, and motivations, on the basis of observation and interpretation. The researcher gives more weight to the views of the participants.

Characteristics:

- Anthropological – Sociological research

- Uncover information
- Natural setting
- Concentrated contact

Types of qualitative research

- Case study
- Grounded theory
- Ethnography
- Historical

Confirmatory research - where researchers have a pretty good idea of what's going on. That is, researcher has a theory (or several theories), and the objective is to find out if the theory is supported by the facts.

Exploratory research - used to establish an understanding of how best to proceed in studying an issue or what methodology would effectively apply to gathering information about the issue.

Characteristics:

- Exploring a phenomenon
- Prior to developing a Ho
- Doesn't test a hypothesis

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Research Questions – basic determining points Two generic

questions are asked in the process:

1. What questions: What phenomena are important?
2. Why questions: Why do some people learn languages better than others?

The WHAT Questions

Characteristics

- Explain phenomenon
- Describe nature
- Description of philosophy

The WHY Questions

After WHAT (Phenomenon + Relation) between variables, 'Why questions' asked

Why questions show causation

Discovering why independent variable affect dependent (at least one)

Used in

- Causal studies
- Relationships and correlations
- Causal comparative studies

- Experimental and quasi experimental

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Extraneous factors - Any variable that you are not intentionally studying in your dissertation is an extraneous variable that could threaten the internal validity of your results.

Confounding variable - When an extraneous variable changes systematically along with the variables that you are studying

Control group may influence in four directions

1. Control group rivalry
2. Experimental treatment diffusion
3. Compensatory equalization of treatments
4. Demoralization of the control group **Five ways the results may be spoiled**

1) Diff instruments used
2) Same instrument for all types of variables
3) The pretest effect
4) The posttest effect
5) Time of measurement effect

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Commonly used data collection procedure in applied linguistics:

1. Observational
2. Instrumental

Observational DGP

- Participant observation
- Non-participant observation
- Sheets
- Judges - raters

Instrumental DGP

- Questionnaires
- Closed-open
- Tests
- Discrete items

Data gathering procedures based on observation

- Self (intro-retro)
- Others – participant (full-non- partial)
- Interviewers
- Judges-raters

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Reliability has to do with the consistency of the data results. If we measure or observe something, we want the method used to give the same results no matter who or what takes the measurement or observations. The indicator used for reporting the reliability is the *correlation coefficient*.

Correlation coefficient is a number that quantifies the degree to which two variables relate to one another.

Reliability coefficients range between 0.00 and +1.00.

- A coefficient of 0.00 would mean that the observations/measurements were inconsistent.
- A coefficient of 1.00 indicates that there is perfect reliability or consistency.

Validity - the ability of an instrument or observational procedure to accurately capture data needed to answer a research question. Types are:

- Content validity
- Predictive validity
- Face validity
- Construct validity

Trait accuracy

The degree to which a procedure is valid for trait accuracy is determined by the degree to which the procedure corresponds to the definition of the trait.

Criterion related simply means that the procedure is validated by being compared to some external criterion. It is divided into two general types of trait accuracy:

- Capacity to succeed - relates to a person having the necessary wherewithal or aptitude to succeed in some other endeavor.
- Current characteristics

Predictive utility is determined by correlating the measurements from the procedures with measurements on the criterion being predicted.

A number of measures have been used over the years to predict the success of students in acquiring a second language. One of the most well-known standardized instruments is the **Modern Language Aptitude Test** developed by Carroll and Sapon.

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Understanding research results

Some people think that numerical data are more scientific—and therefore more important—than verbal data. We must not forget that when we use statistics, we have basically transferred verbally defined constructs into numbers so we can analyze the data more easily.

We must not forget that these statistical results must again be transferred back into terminology that represents these verbal constructs to make any sense.

Verbal data commonly appear as selections of excerpts, narrative vignettes, and quotations from interviews

Numerical data are often condensed into tables of frequencies, averages

The researcher has at least five strategies to choose from to support the quality of the data

1) Representativeness - This is not referring specifically to whether the sample is representative of the population.
2) Researcher effects - looking at the unidirectional effect of the researcher on the behavior of the persons from which data were being collected
3) Prolonged engagement and persistent observation: The researcher needs enough time to interact with the respondents
4) Clarifying researcher bias - the researcher should disclose any biases that may have an impact on the approach used
5) Weighting the evidence - some data are stronger (or more valid) than others

Evaluating explanations and conclusion

Spurious relationships: Not all things that appear to be related are directly related. For example, lung cancer and the number of ashtrays a person owns are related. However, this relationship is spurious

If–then tests: The researcher tests his or her hypothesized explanation by predicting that some consequent would occur with a novel sample of people or set of events.

Rival explanations: Eliminating competing explanations is a powerful way to add weight to a theoretical conclusion.

Replicating findings: The more often the same findings occur despite different samples and conditions, the more confidence we can have in the conclusions.

Informant feedback: This relates to the reactions that the informants have to the conclusions of the study. Such feedback can be used to check the plausibility of patterns perceived by the researcher.

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Understanding statistics of data

There are three basic concerns that should be addressed when using descriptive statistics to describe numerical data:

1) The shape of the distribution - The concern is whether the data are symmetrically distributed and approximate a normal curve.
2) Measures of average - concern is which statistic to use to describe average. There are 3: mean, median and mode
3) Measures of variation - concern is what statistic to use to indicate how much the data vary. There are 3 measures of variation:
   - Standard deviation
   - Semi-interquartile range
   - Range

Inferential statistics can be divided into two general categories:

1) Parametric - statistics are used for analyzing data in the form of frequencies, ranked data
2) Non-parametric - statistics are used for any data that do not stray too far from a normal distribution and typically involve the use of means and SD

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Discussion

- An overview of the study
- Overview of the findings - how the findings address the research question
- Relation of findings - The researcher should relate the findings of his or her study to previous research findings
- Attention to limitations - point out any weaknesses and/or limitations regarding the study.
- Possible applications - how the results can be applied to practical situations.
- Future possibilities - The researcher should suggest topics for future research. **Discussion vs. conclusion**

- D: Interpret results
- D: Compare with previous
- D: Discuss limitations
- D: Unexpected results?
- D: How they add value?
- C: Restate your hypothesis
- C: Important findings
- C: Highlight limitations
- C: Overall significance
- C: Future directions

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